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TransitFlow: flow-matching posterior estimation for candidate-conditioned exoplanet transit inference

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ABSTRACT

We present TransitFlow, a candidate-conditioned simulation-based inference pipeline that combines a transit-candidate probability with a flow-matching posterior over $R_\mathrm{p}/R_\star$, $a/R_\star$, impact parameter b, and two quadratic limb-darkening parameters. The method consumes global, candidate-folded local, and box-periodogram views of a single TESS-like light curve. A diagnostic single-seed experiment gives pooled simulator-domain simulation-based calibration and a mean absolute coverage error of 0.015, but conditional 68 per cent coverage for b decreases to 0.564 in the high-b stratum. We identify and correct an evaluation design in which the neural classifier received the simulated true ephemeris while Box Least Squares (BLS) had to search. In a corrected $n = 1000$ candidate-vetting experiment, TransitFlow achieves ROC-AUC 0.807 (95 per cent paired-bootstrap interval 0.782–0.834) and average precision 0.762 (0.731–0.796), compared with 0.598 and 0.584 for the BLS score. This result is preliminary because the real-noise library is not yet separated by target and sector. A selected positive-only sample recovers 30 known TESS transits, but does not establish real-world reliability, and previous real-posterior agreement values are invalidated by collapsed importance-sampling effective sample size and unconverged MCMC references. TransitFlow is therefore a promising candidate-conditioned inference method, not yet a validated blind search or real-characterization system. We specify the frozen experiments needed for an MNRAS-ready result.

Key words: methods: data analysis – methods: statistical – planets and satellites: detection – techniques: photometric

1 INTRODUCTION

Transit surveys require both candidate discovery and physical interpretation. Box Least Squares (BLS; Kovács et al. 2002) and Transit Least Squares (TLS; Hippke & Heller 2019) provide effective periodic-search statistics, whereas learned vetters such as AstroNet (Shallue & Vanderburg 2018) and ExoMiner (Valizadegan et al. 2022) classify threshold-crossing events from richer data products. Those systems are primarily optimized for discrimination. Physical transit parameters and their uncertainties are normally inferred in a second stage using likelihood-based sampling; joint Bayesian detection and noise characterization has also been explored
citepTaaki2020.

Simulation-based inference (SBI) offers a complementary route when a forward simulator is available but repeated posterior sampling is expensive (Cranmer et al. 2020; Lueckmann et al. 2021). Neural posterior estimators amortize the cost of inference over simulations. Their scientific utility nevertheless depends on calibration and fidelity rather than predictive accuracy alone; apparently plausible neural posteriors can be unfaithful (Hermans et al. 2022), motivating simulation-based calibration (SBC; Talts et al. 2018), expected-coverage tests, posterior predictive checks and explicit out-of-distribution diagnostics.

Flow-matching posterior estimation (FMPE) applies continuous flow matching (Lipman et al. 2023) to SBI (Dax et al. 2023). It has been demonstrated for exoplanet atmospheric retrieval (Gebhard et al. 2025). Conditional flow matching has also been used for transit-light-curve classification (Fiscale et al. 2025). TransitFlow therefore does not claim the first use of flow matching for transit detection, nor a new flow-matching algorithm. Its intended contribution is narrower: an FMPE posterior for candidate-conditioned transit photometry, coupled to a candidate probability and evaluated with calibration and misspecification diagnostics.

This paper has two aims. First, we define the model and the astronomical task without conflating candidate vetting with blind search. Secondly, we audit the available evidence under that definition. The audit uncovered privileged ephemeris information in the original detection benchmark, a non-independent noise evaluation, and invalid importance-resampled real-posterior comparisons. We correct the corresponding code paths and report only the results that remain interpretable. The resulting manuscript is deliberately a pre-submission draft: Section 7 states the experiments that must be completed before the method can support an MNRAS submission.

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2 PROBLEM FORMULATION

Let x denote a single-sector light curve and $c = (P, t_0)$ a proposed candidate ephemeris. TransitFlow models the factorization

$$p(d, \boldsymbol{\theta} \mid x, c) = p(d \mid x, c)\, p(\boldsymbol{\theta} \mid d = 1, x, c), \tag{1}$$

where d indicates whether the candidate is transit-like and

$$\boldsymbol{\theta} = (R_{\rm p}/R_\star, a/R_\star, b, q_1, q_2). \tag{2}$$

The period and epoch are conditioning variables rather than posterior targets. This distinction is important: posterior plots or coverage statistics for P and t_0 would not measure characterization calibration in the present five-dimensional model.

The factorization is implemented by two heads sharing an observation embedding (Fig. 1). It is a practical joint interface, but not a single mixed discrete–continuous posterior. At deployment, BLS or TLS must propose c. A benchmark that supplies the true positive ephemeris measures an oracle-candidate diagnostic, not end-to-end discovery.

3 FORWARD MODEL AND NEURAL POSTERIOR

3.1 Simulator and priors

The forward simulator produces 27-d TESS-like light curves with 18000 raw cadences. Planet parameters are sampled from bounded priors. The period lies between 0.5 and 13 d; $R_{\rm p}/R_\star$ is log-uniform over 0.01–0.15; $a/R_\star$ is induced by a stellar-density prior; and grazing configurations are included through the support of b. Quadratic limb darkening is sampled in the physical (q_1, q_2) parameterization of Kipping (2013). Transit curves are exposure-integrated using a native implementation checked against BATMAN (Kreidberg 2015).

The noise mixture assigns 70 per cent of examples to cached out-of-transit TESS segments, 20 per cent to Matérn-3/2 correlated noise plus white noise, and 10 per cent to white noise. Negative examples include eclipsing-binary-like dips, isolated events and sinusoids. Dilution, cadence gaps and finite exposure are simulated. A transit-preserving median filter removes slow trends. In the current archive, 115 real-noise segments generated the training data and the same library was reused for the diagnostic synthetic evaluation. Because the archive does not retain target and sector identifiers, this reuse cannot be repaired retrospectively; a grouped rebuild is required.

3.2 Representation and network

Each curve is represented by a 2001-bin global view, a 201-bin local view folded on c, and a 256-bin box-periodogram view. A tri-branch residual convolutional network maps the views, a noise-level feature, the standardized candidate ephemeris and a dilution feature to a 256-dimensional embedding. A binary head returns $p(d = 1 \mid x, c)$. The FMPE head defines a conditional time-dependent velocity field in standardized parameter space and is trained with an optimal-transport conditional flow-matching objective,

$$\mathcal{L}_{\rm FM} = \mathbb{E}\, \| v_\phi(t, \boldsymbol{z}_t \mid x, c) - (\boldsymbol{z}_1 - \boldsymbol{z}_0) \|_2^2 \,, \tag{3}$$

where $\boldsymbol{z}_0 \sim \mathcal{N}(0, I)$ and $\boldsymbol{z}_1$ is the standardized simulated truth. At inference, an ODE transports base samples to posterior samples. The evaluated model contains approximately 4.18 million parameters and was trained for 60000 steps on one million simulated curves.

4 EVALUATION DESIGN

4.1 Simulator-domain calibration

The archived diagnostic uses 1000 simulated planets and 2000 posterior samples per object. Per-parameter SBC rank histograms are tested for uniformity with a Bonferroni familywise threshold of 0.05 across the five characterization parameters. Passing such a test means that miscalibration was not detected; it does not prove calibration. Expected coverage is measured over central credible intervals from 5 to 95 per cent. Coverage is also stratified by b, $R_{\rm p}/R_\star$, $a/R_\star$, signal-to-noise ratio and number of observed transits.

This evaluation is useful for diagnosing the trained network but is not the final held-out test because the real-noise segment groups overlap training. The final protocol will split target–sector groups before simulation and freeze the test library until all model and threshold choices are complete.

4.2 Candidate-vetting comparison

The original comparison constructed the neural local view from the true positive ephemeris while requiring BLS and TLS to search the raw light curve. That design answers different questions for the compared methods. We replace it with a pipeline in which BLS searches each unflattened curve; its best period, epoch and duration then define TransitFlow's preprocessing and local view. The BLS score and TransitFlow probability are evaluated on identical examples. Uncertainty is estimated with a stratified paired percentile bootstrap that resamples positive and negative examples separately. The current corrected experiment uses $n = 1000$ balanced examples and 1000 bootstrap replicates.

4.3 Real-light-curve checks

The archived real sample consists of 30 confirmed TESS planets chosen inside the training support and subjected to cadence, signal-to-noise and geometry quality cuts. The published period and a data-refined epoch define the candidate. This is a positive-only recovery study. It cannot estimate real specificity, false-positive rate, precision or ROC-AUC.

A previous comparison used EMCEE (Foreman-Mackey et al. 2013) on 16 objects with the ephemeris fixed. Importance weights were then used to resample the amortized posterior before Wasserstein distances were calculated. The median ESS fraction was 0.00181 and the minimum 0.000333, and all chains were shorter than 50 estimated autocorrelation times. Those corrected Wasserstein values are not scientifically interpretable. The implementation now reports raw amortized agreement independently, records correction diagnostics without overwriting the raw result, seeds the independent EMCEE proposal state, and gates the reference posterior on at least 50 autocorrelation times and 400 effective samples.

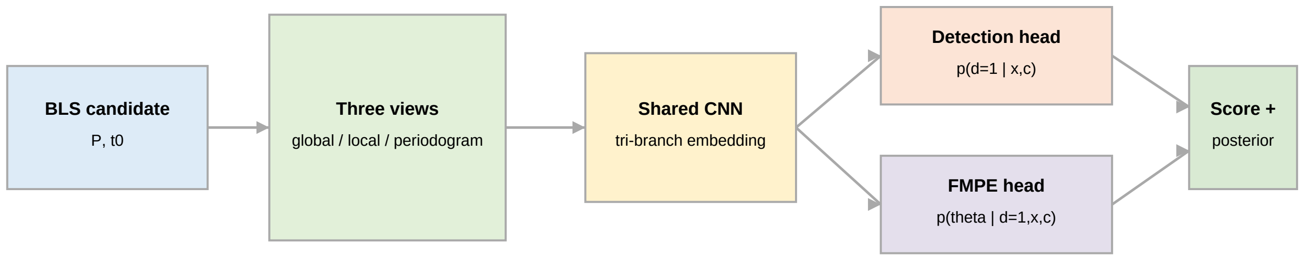

Figure 1. TransitFlow's candidate-conditioned task. A search algorithm proposes a period and epoch. Global, candidate-folded local and periodogram views feed a shared embedding; separate heads produce a candidate probability and an FMPE posterior over five transit-shape parameters. The candidate generator is part of any blind-search evaluation.

Table 1. Corrected BLS-candidate comparison on 1000 simulations. Intervals are paired stratified bootstrap 95 per cent intervals.

Method	ROC-AUC	Average precision
BLS score	0.598 [0.561, 0.631]	0.584 [0.551, 0.620]
TransitFlow	0.807 [0.782, 0.834]	0.762 [0.731, 0.796]
Paired gain	0.209 [0.166, 0.255]	0.178 [0.128, 0.226]

5 DIAGNOSTIC RESULTS

5.1 Pooled and conditional calibration

The single archived run has a minimum characterization SBC p-value of 0.107 and a pooled mean absolute expected-coverage error of 0.0153. Figure 2 shows that the pooled curve is close to the diagonal while conditional behaviour is weaker. The 68 per cent coverage for b is about 0.64, 0.68 and 0.564 in the low-, intermediate- and high-b strata, respectively. The high-b result is inconsistent with nominal coverage at the available sample size. Consequently, the manuscript must not state that the posterior is calibrated uniformly across the prior.

5.2 Fair candidate-vetting pilot

Table 1 and Fig. 3 report the corrected candidate-level experiment. TransitFlow improves paired ROC-AUC over the BLS score by 0.209 (95 per cent bootstrap interval 0.166–0.255) and AP by 0.178 (0.128–0.226). The absolute ROC-AUC of 0.807 is substantially below the historical oracle-candidate value of 0.9993 on the same seeded 1000-example simulation. BLS recovers the injected period to within 1 per cent for only 32.0 per cent of planets under the deliberately coarse 200-period search; the median direct fractional error is near unity because harmonics are frequent. The result shows both that candidate conditioning is useful and that candidate generation is a dominant part of the deployed task.

5.3 Real recovery and posterior status

The selected positive sample gives candidate probabilities above 0.9 for all 30 known transits. The two-sided exact 95 per cent lower confidence bound for a 30/30 binomial result is approximately 0.884, even before selection effects are considered. The result is evidence that the network responds to high-quality known candidates, not evidence for catalogue-level reliability.

No real-characterization agreement number is promoted to a paper result in this draft. The archived point estimate for the prior-normalized b Wasserstein distance fluctuates across the predeclared 0.1 threshold among adjacent runs, and the reported corrected samples have collapsed ESS. A rerun using the raw amortized posterior and a converged, matched reference is mandatory.

6 DISCUSSION

The corrected experiment changes the interpretation of TransitFlow. The method is best viewed as an amortized candidate vetter and characterization engine downstream of a search stage. The search determines whether the local view is coherent; training only on true positive ephemerides and random negative ephemerides creates an avoidable train–deployment mismatch. Future training must include BLS/TLS candidates, aliases, perturbed epochs and near-miss candidates. The large oracle-to-deployed AUC change makes candidate robustness a primary experiment, not an appendix.

The conditional coverage result also illustrates why pooled SBI diagnostics are insufficient. Impact parameter is degenerate with scaled semimajor axis and limb darkening in single-sector photometry. Stratified coverage, posterior predictive checks and held-out real-noise groups are needed to distinguish estimator error from simulator mismatch. Importance correction can diagnose missing support, but an ESS of one or a few effective draws cannot repair the posterior.

The current speed measurement is likewise provisional. Amortized inference is expected to outperform per-object MCMC after training, but a scientific speedup must compare methods at matched accuracy and convergence on the same hardware, include preprocessing and candidate-search cost, and state the amortization break-even point. The archived 1659.7-fold figure does not meet all these conditions.

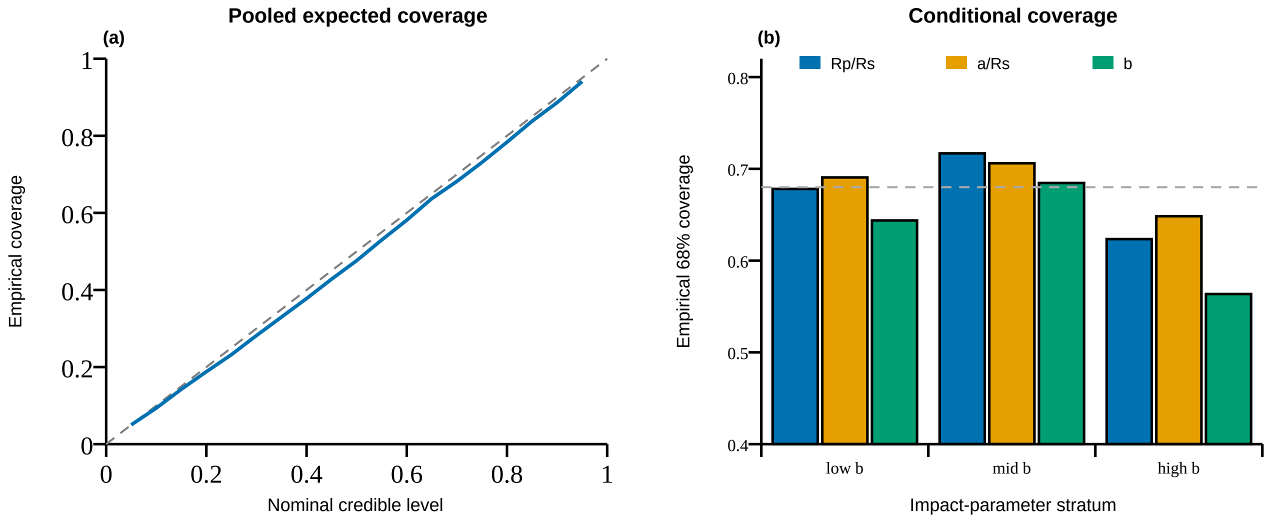

Figure 2. Simulator-domain calibration diagnostic for the archived single-seed run. (a) Pooled expected coverage is close to the ideal diagonal. (b) Empirical 68 per cent coverage stratified by impact parameter; the dashed line denotes nominal coverage. The decrease for b at high impact parameter is hidden by the pooled summary. This figure does not constitute an independent held-out result because the real-noise segment library overlaps training.

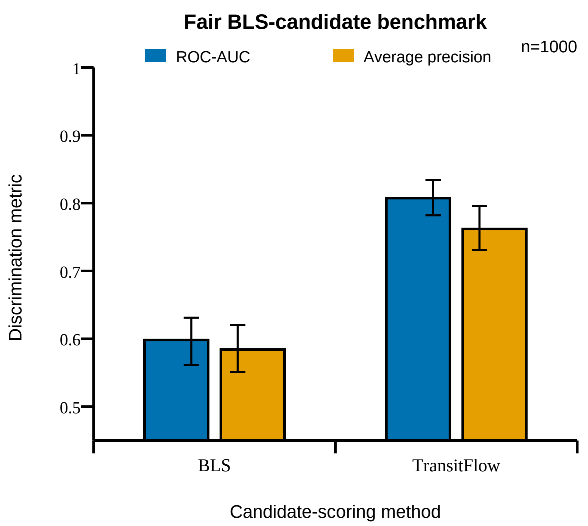

Figure 3. BLS and TransitFlow discrimination when both operate in the same BLS-candidate pipeline. Error bars are paired-bootstrap 95 per cent intervals. These simulations reuse the training noise library and are therefore a pilot, not the final held-out benchmark.

7 BLOCKING VALIDATION BEFORE SUBMISSION

The following items are preregistered as submission gates for the next frozen run.

(i) Split real-noise segments by target and sector before any injection; retain immutable train, validation and test manifests.

(ii) Train at least three independent seeds and report seed-level outcomes, means, standard deviations and gate flips.

(iii) Evaluate BLS, TLS and TransitFlow on the same held-out examples and candidate task, with paired confidence intervals, reliability diagrams and a modern learned vetter baseline.

(iv) Freeze an untouched real sample containing confirmed planets and matched astrophysical/instrumental false positives, with exact MAST product identifiers.

(v) Compare raw amortized posteriors against matched MCMC or nested-sampling references using multiple chains, split-$\hat{R} < 1.01$, bulk and tail ESS above 400, and at least 50 autocorrelation times.

(vi) Add FMPE–NPE, noise-regime and candidate-perturbation ablations, and conditional coverage with simultaneous uncertainty bands.

(vii) Validate BF16 against FP32 on the same checkpoint, tensors, base draws and hardware with predeclared tolerances.

(viii) Repeat timings only after convergence and accuracy are matched.

The manuscript must remain marked as a pre-submission draft until every gate is resolved or the corresponding scientific claim is removed.

8 CONCLUSIONS

TransitFlow implements flow-matching posterior estimation for candidate-conditioned transit photometry and exposes a useful factorized interface for candidate scoring and physical posterior sampling. Its pooled simulator-domain diagnostics are promising, and a corrected candidate-level pilot shows a paired improvement over the BLS score. The same audit also reveals conditional high-b miscalibration, strong dependence on candidate quality, non-independent noise evaluation, a positive-only real sample and invalid old importance-corrected real-posterior comparisons. The scientifically honest conclusion is therefore narrower than the original repository narrative: TransitFlow is a promising research prototype whose MNRAS case depends on a new, frozen and leakage-safe validation, not on the current full-run label.

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AUTHOR CONTRIBUTIONS

CRediT roles and the final author list must be supplied and approved by all human authors before submission. Artificial-intelligence systems are not authors.

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CONFLICT OF INTEREST

The human authors must confirm the final conflict-of-interest statement before submission. No conflict statement is inferred automatically.

AI-USE DISCLOSURE

OpenAI Codex was used as an assistive tool to inspect code and result artifacts, identify reproducibility defects, implement and test software changes, draft text, and generate figure code. All scientific claims, citations, analyses and submission materials require verification and approval by the human authors. This use must also be disclosed in the cover letter, in accordance with MNRAS instructions.

DATA AVAILABILITY

The development source and diagnostic result artifacts are currently available in the project repository. Before submission, the exact executed source, environment, grouped data manifests, plotting tables, commands, checkpoint and noise-library hashes, and final result bundles will be deposited in an immutable public archive with a DOI. TESS light curves are available from MAST and catalogue metadata from the NASA Exoplanet Archive; the final release will provide exact product identifiers and access dates. The present draft has no archival DOI and is not independently reproducible from the repository alone because the retained 115-segment legacy noise library lacks source-target identifiers and therefore cannot support a leakage-safe grouped evaluation.

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APPENDIX A: REPRODUCIBILITY FIELDS FOR THE FROZEN RUN

The final machine-readable manifest will include the Git commit and patch hash, configuration, random seeds, package lock, Python/PyTorch/CUDA/cuDNN versions, GPU model, target–sector group manifests, MAST identifiers, simulator and model hyperparameters, primary checkpoint hash, per-object scores/posteriors, MCMC diagnostics and all plotting data. A gate report that omits failed diagnostics is not sufficient provenance.